



Hepatitis Viruses — Hepatitis A and E

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which hepatitis viruses are enterically (feco-orally) transmitted, and which are parenteral?
2. Which hepatitis viruses can become chronic, and which never do?
3. What makes Hepatitis E uniquely dangerous in pregnant women, unlike Hepatitis A?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 What is hepatitis?

- **Hepatitis** = inflammation of the liver (*hepat* = liver + *itis* = inflammation)
- **Viral hepatitis** = a specific virus causing the liver to inflame (swell/enlarge)

Frequency	Causes of viral hepatitis
Common	Hepatitis A, B (± D), C, E
Less common	Cytomegalovirus, Epstein-Barr virus
Rare	Herpes simplex, Yellow fever

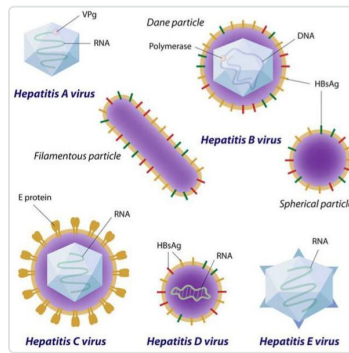
Presentation	Detail
Non-specific (most common)	Fever, headache, myalgia, arthralgia, loss of appetite — frequently anicteric or even asymptomatic
Liver-specific (fewer patients)	Pale stool, dark urine, jaundice, abdominal pain, mild hepatosplenomegaly, weight loss
Classic picture	Mild jaundice + abnormal liver blood tests with nonspecific symptoms

2 Comparing the hepatitis viruses

This lecture develops **Hepatitis A and Hepatitis E in full clinical detail** (see sections 3–4). Hepatitis B, C, D, F, and G appear only in the comparison tables below — they are named and classified here but not covered with dedicated pathogenesis/clinical/diagnosis content in this deck.

Virus	Family	Genome	Structure
Hepatitis A (HAV)	Picornaviridae	ssRNA	Naked
Hepatitis B (HBV)	Hepadnaviridae	dsDNA	Enveloped
Hepatitis C (HCV)	Flaviviridae	ssRNA	Enveloped
Hepatitis D (HDV)	Delta virus (defective; needs HBV)	ssRNA	Enveloped
Hepatitis E (HEV)	Hepeviridae	ssRNA	Naked
Hepatitis F (HFV)	mutant of HBV	—	—
Hepatitis G (HGV)	Flaviviridae	ssRNA	—

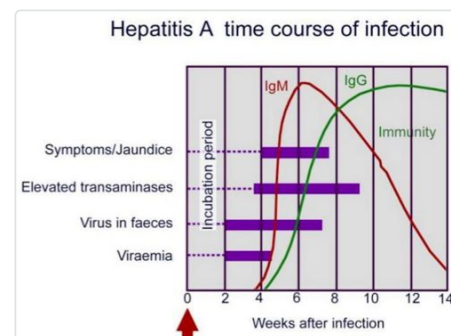
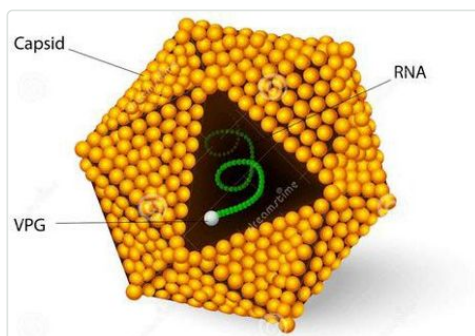
Virus	Transmission	Stability	Incubation (weeks)	Prevalence	Chronicity	Oncogenicity
HAV	Enteric (feco-oral)	Heat & acid stable	2–4	High	Never	Non-oncogenic
HBV	Parenteral	Acid sensitive	4–20	High	Often	Oncogenic
HCV	Parenteral	Acid sensitive	2–26	Moderate	Often	Oncogenic
HDV	Parenteral	Acid sensitive	6–9	Low/regional	Often	Unknown
HEV	Enteric (feco-oral)	Heat & acid stable	3–8	Regional	Never	Non-oncogenic



Structure of the five hepatitis viruses.

3 Hepatitis A — infectious hepatitis

- **Picornavirus** group of enteroviruses; **highly infectious**; **feco-oral** spread
- **Common in overcrowding and poor sanitation**
- The virus is **resistant to heat and chemicals**
- **Benign disease with complete recovery** within several weeks; **immunity after infection is usually lifelong**



HAV capsid — icosahedral, naked, with VPg and RNA genome. Time course of HAV infection: symptoms/jaundice, transaminases, fecal virus shedding, and viremia relative to IgM/IgG response.

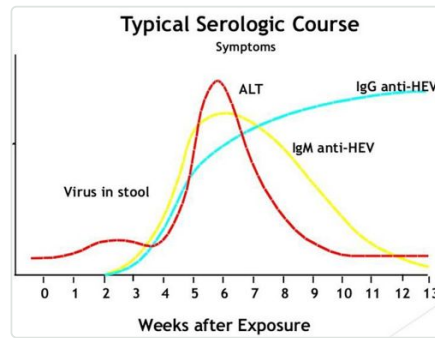
Outcome (%)	Children	Adults
Inapparent (subclinical) infection	80-95	10-25
Icteric disease	5-20	75-90
Complete recovery	>98	>98
Chronic disease	None	None
Mortality rate	0.1	0.3-2.1

Key age pattern: children are mostly **asymptomatic/subclinical**, while adults are far more likely to develop **icteric (jaundiced) disease** — though both groups have >98% complete recovery and no chronic carriage.

	Detail
Diagnosis	Serology — anti-HAV IgM (acute/current infection) vs IgG (past infection/immunity)
Prevention	Better personal hygiene; passive immunization (immunoglobulin); active immunization (vaccine)
Treatment	No specific antiviral treatment — supportive care only

4 Hepatitis E — enteric non-A, non-B hepatitis

- Family **Hepeviridae**; **feco-oral transmission**
- **Endemic in India and the Middle East**
- Causes an **acute, self-limiting infection with low mortality** in the general population
- Clinical presentation and management are **similar to Hepatitis A**
- **Chronic carrier state does not occur**



Typical HEV serologic course: ALT rise, fecal virus shedding, then IgM anti-HEV followed by IgG anti-HEV.

Key difference from Hepatitis A: HEV infection during **pregnancy** is associated with **acute liver failure and high mortality** — a risk not shared by HAV.

	Detail
Diagnosis	Serology — anti-HEV IgM
Prevention	Better personal hygiene; avoiding contaminated seafood
Treatment	Supportive

P Must-remember pearls

- **Enteric (fecal-oral):** HAV, HEV — both **naked, heat/acid-stable, never chronic, non-oncogenic**.
Parenteral: HBV, HCV, HDV — all **enveloped, acid-sensitive, often chronic**; HBV and HCV are **oncogenic**.
- **HAV:** children mostly subclinical, adults mostly icteric; **>98% complete recovery either way**, immunity usually lifelong after infection.
- **HEV in pregnancy = acute liver failure with high mortality** — the single biggest HAV-vs-HEV exam differentiator.
- Diagnosis for both HAV and HEV is **serology: IgM = acute infection, IgG = past infection/immunity**.
- Neither HAV nor HEV has a specific antiviral — management is **supportive**; prevention = hygiene ± immunization (HAV has both passive and active vaccination options).

Q Test yourself

1. A middle-aged man has acute fever, nausea, right upper quadrant pain, jaundice, and dark urine. Anti-HAV IgM is positive. Can he transmit infection to family members, and for how long is this classically described?

- A. Yes, by person-to-person feco-oral spread, for up to about 2 weeks
 B. No, HAV cannot spread person-to-person
 C. Yes, but only via blood transfusion
 D. Yes, indefinitely as a chronic carrier

2. What is the genome type of Hepatitis A virus?

- A. Single-stranded RNA (ssRNA)
 B. Double-stranded DNA
 C. Single-stranded DNA
 D. Double-stranded RNA

3. Which hepatitis virus uniquely causes acute liver failure with high mortality when infection occurs during pregnancy?

- A. Hepatitis E virus (HEV)
 B. Hepatitis A virus (HAV)
 C. Hepatitis C virus (HCV)
 D. Hepatitis D virus (HDV)

4. A child and an adult are both infected with HAV. Which outcome pattern is expected?

- A. The child is more likely to be asymptomatic/subclinical; the adult is more likely to develop icteric disease
 B. Both are equally likely to develop icteric disease
 C. The child is more likely to develop chronic hepatitis
 D. The adult is more likely to be asymptomatic

5. What is the mainstay of management for both Hepatitis A and Hepatitis E infection?

- A. Supportive care, as there is no specific antiviral treatment
 B. Direct-acting antivirals
 C. Interferon plus ribavirin
 D. Immediate liver transplantation in all cases

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — HAV spreads feco-orally person-to-person, with transmission risk to close contacts for roughly 2 weeks around the acute illness.
- 2 — A** — HAV, a picornavirus, has a single-stranded RNA genome.
- 3 — A** — HEV infection during pregnancy is specifically associated with acute liver failure and high mortality, unlike HAV.
- 4 — A** — In HAV, children have 80-95% inapparent infection while adults have 75-90% icteric disease — age strongly shifts presentation.
- 5 — A** — Neither HAV nor HEV has a specific antiviral treatment; management for both is supportive.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

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Study alongside the lecture, not instead of it.